

Exploring the Effect of Personalized Virtual Reality Serious Game for Stroke Rehabilitation



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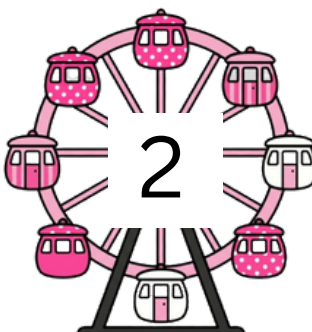
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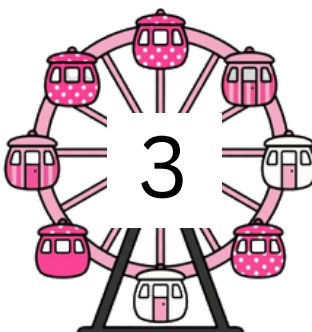
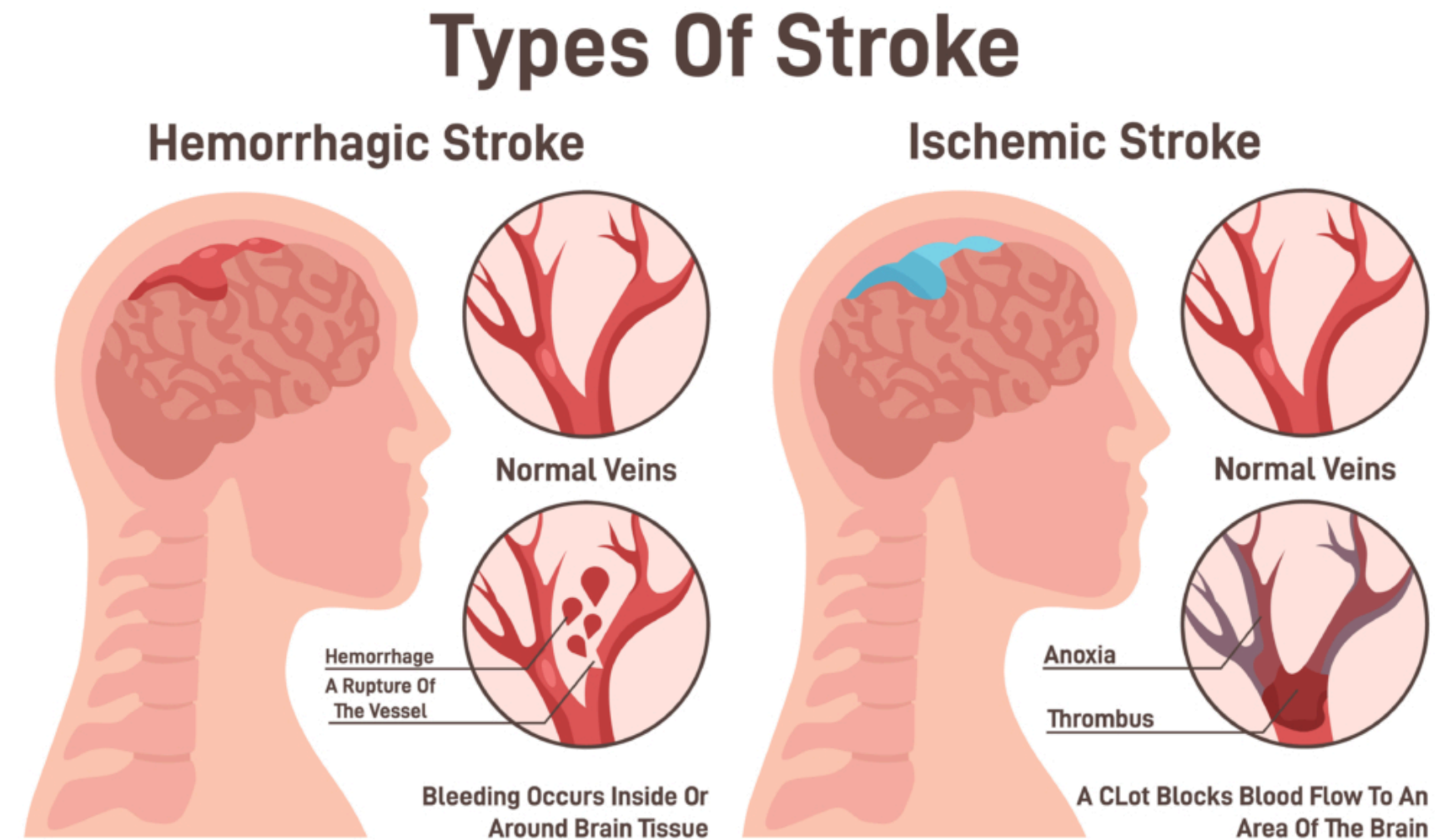
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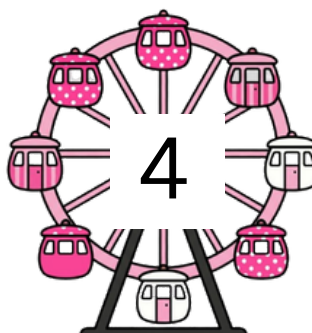
Introduction Stroke

- 12M+ new cases/year worldwide
- 94M+ people living with stroke consequences
- Impairments: mobility, cognition, speech & vision
- Early rehabilitation is key to maximizing recovery



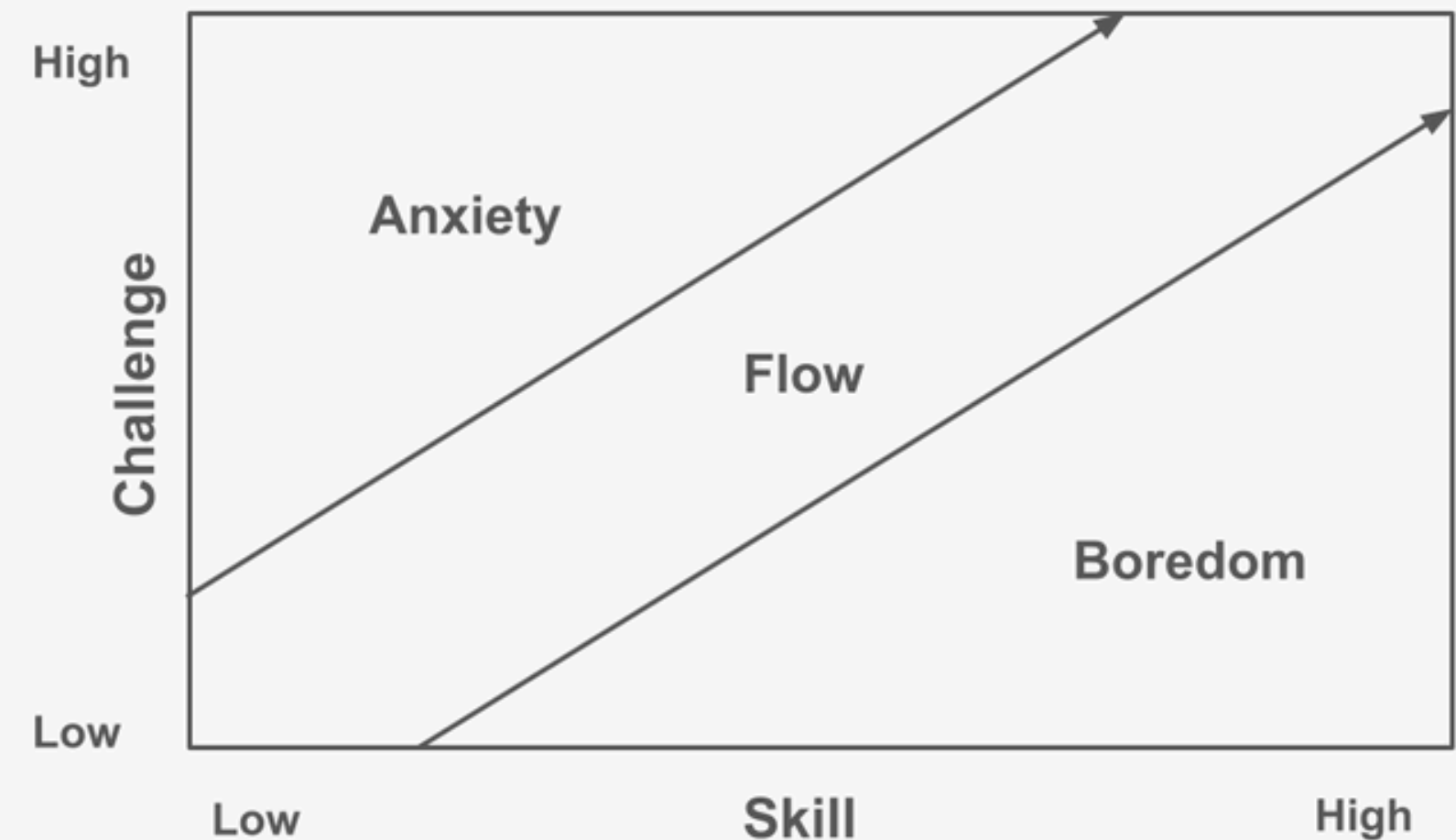
Limitations of Conventional Rehabilitation

- Key limitation: monotony and lack of individualization
- Low motivation leads to poor adherence and recovery failure

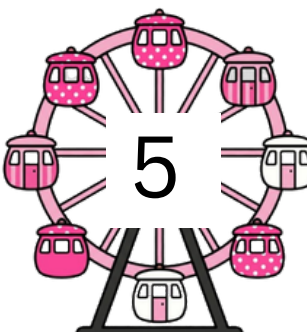


The Importance of Personalization

- Adapts to survivors' individual needs
- Dual benefit: greater adherence, sustained neuroplasticity
- Real-time challenge adaptation (Flow Zone)

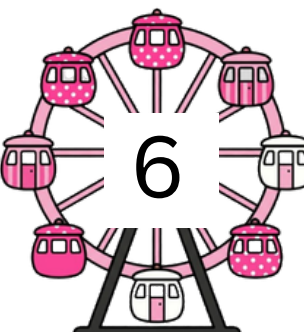


- Csikszentmihalyi (1990), Flow: The Psychology of Optimal Experience



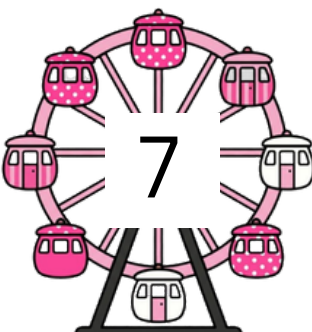
Objectives

- Develop a personalized VR serious game for stroke rehabilitation
- Implement an automatic rule-based algorithm
- Apply HCD methodology
- Evaluate usability, motivation & personalization perceptibility



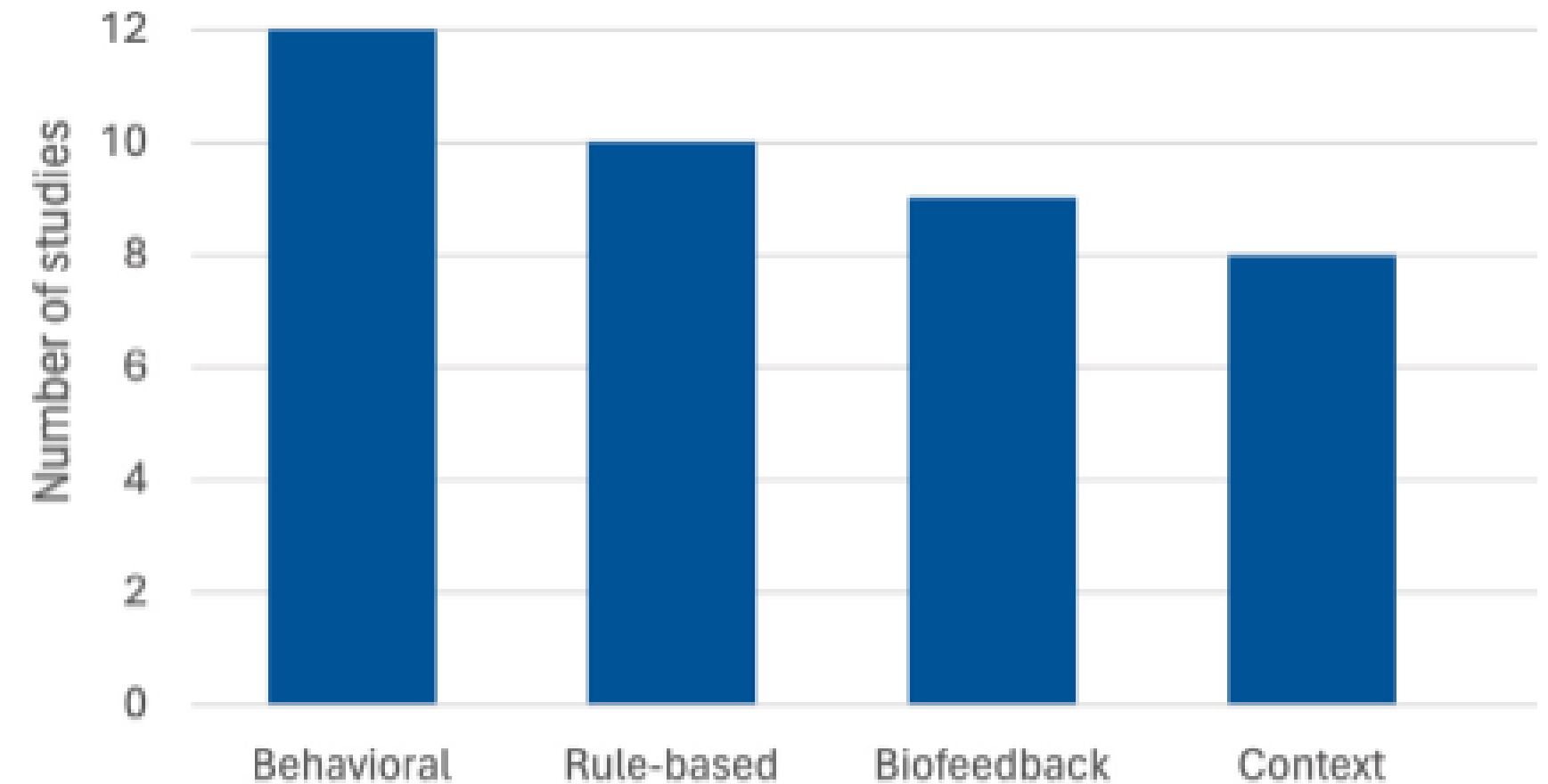
Background

- VR in 84.6% of studies, Serious Games in 67%
- Hand-tracking preferred: 41% vs 25.6% (controllers)
- Main focus: upper limb rehabilitation (66.7%)
- Gap: most physiotherapists have yet to adopt VR clinically

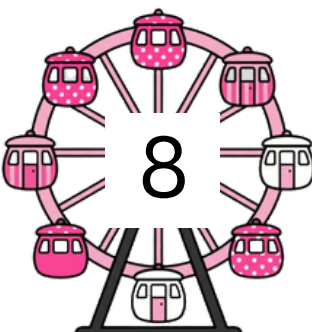


Background

- Personalization strategies in VR: Behavioral Tracking, Rule-based, Biofeedback, Context adaptation
- Automatic personalization dominates literature but not clinical preference

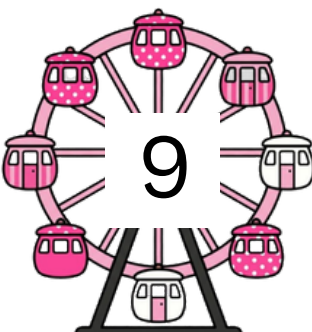
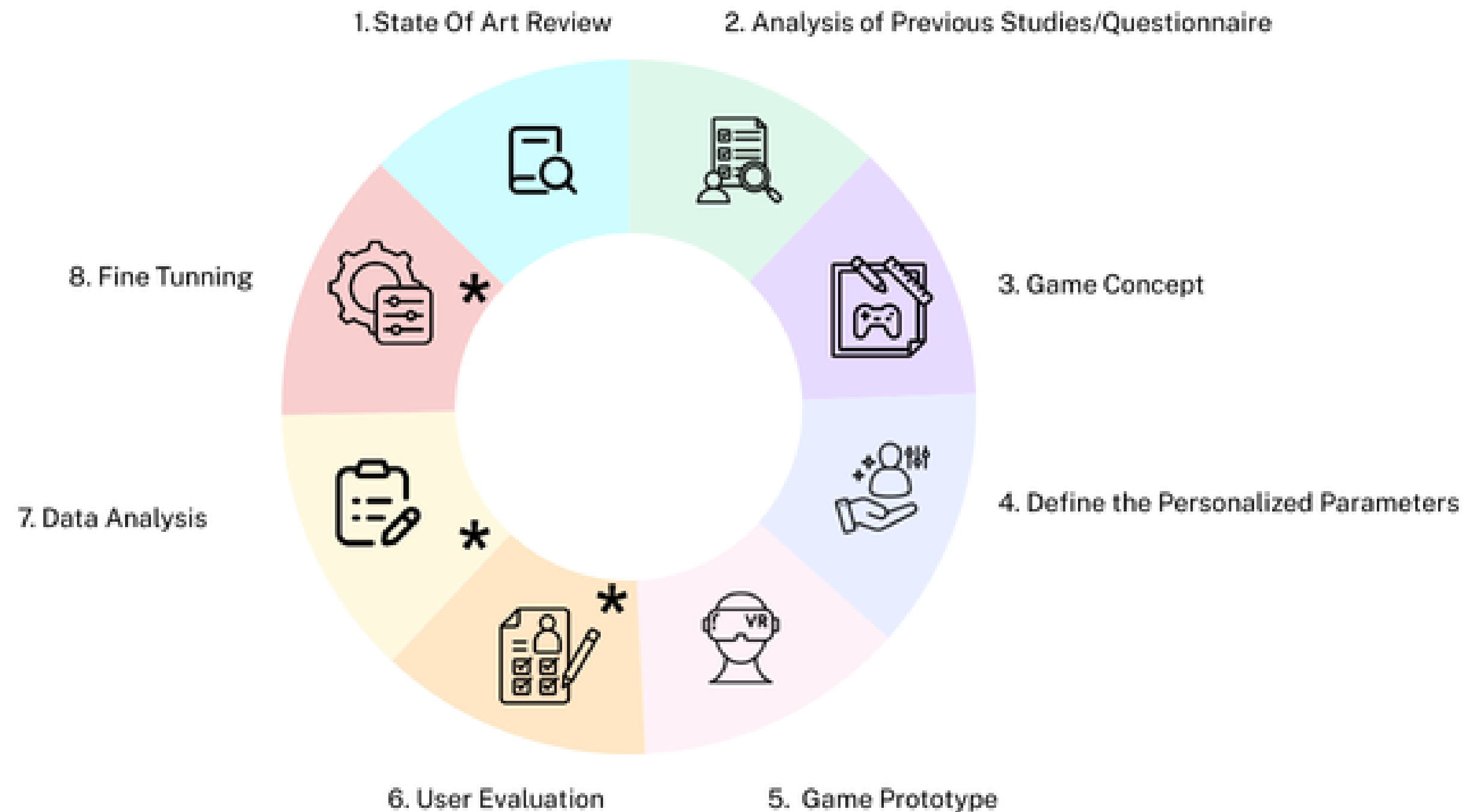


- Figueiredo et al. (2025), Personalized extended reality experiences to enhance the rehabilitation process of stroke survivors: A scoping review

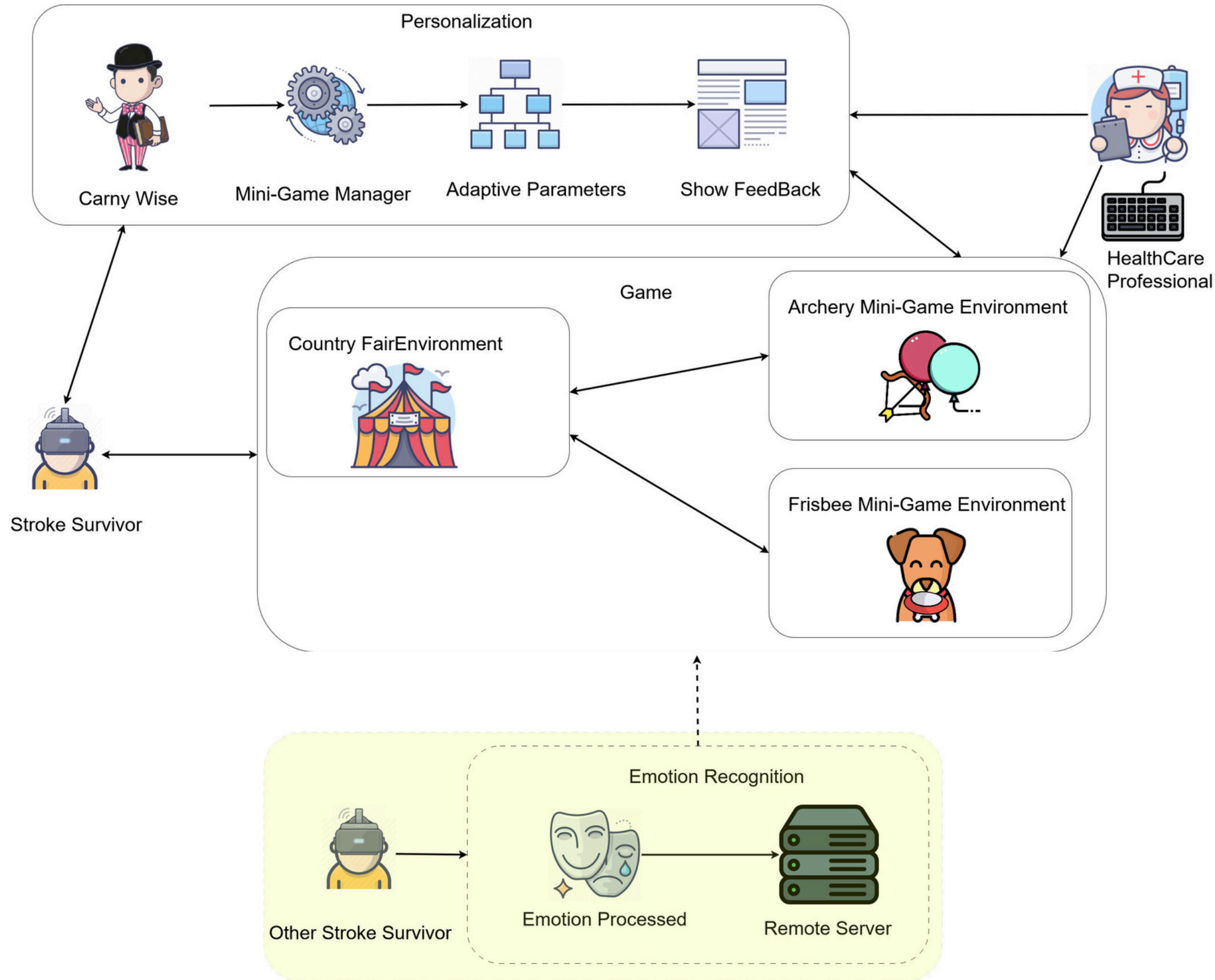
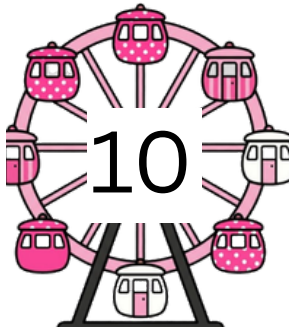
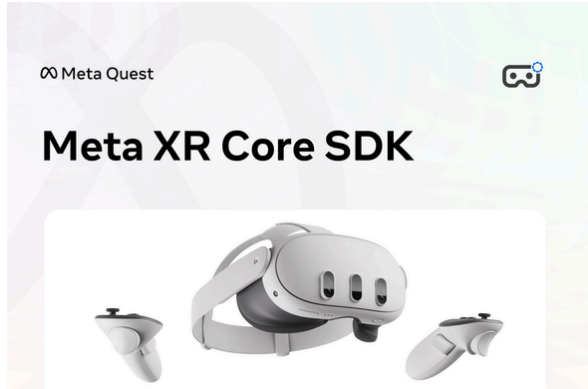


Methodology

Human Centered Design



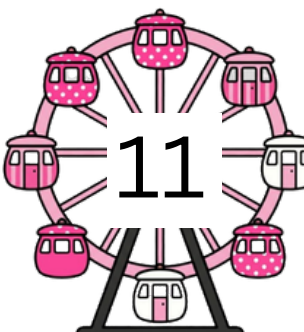
System Architecture



Country Fair Environment



- Figueiredo et al. (2026): Addressing Personalized Virtual Reality for Stroke Rehabilitation Through a Country Fair Serious Game Approach: A Proposal, IEEE VRW



Archery Mini-Game

Motor

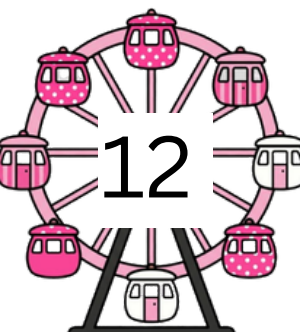
- Virtual bow to pop colored balloons
- grab gesture
- Activates biceps, latissimus dorsi & triceps

Cognitive

- Dual-task: aiming and color rule (attention & inhibition)

Personalization Parameters

- Balloon quantity, density, movement, transparency and color rule



Frisbee Mini-Game

Motor

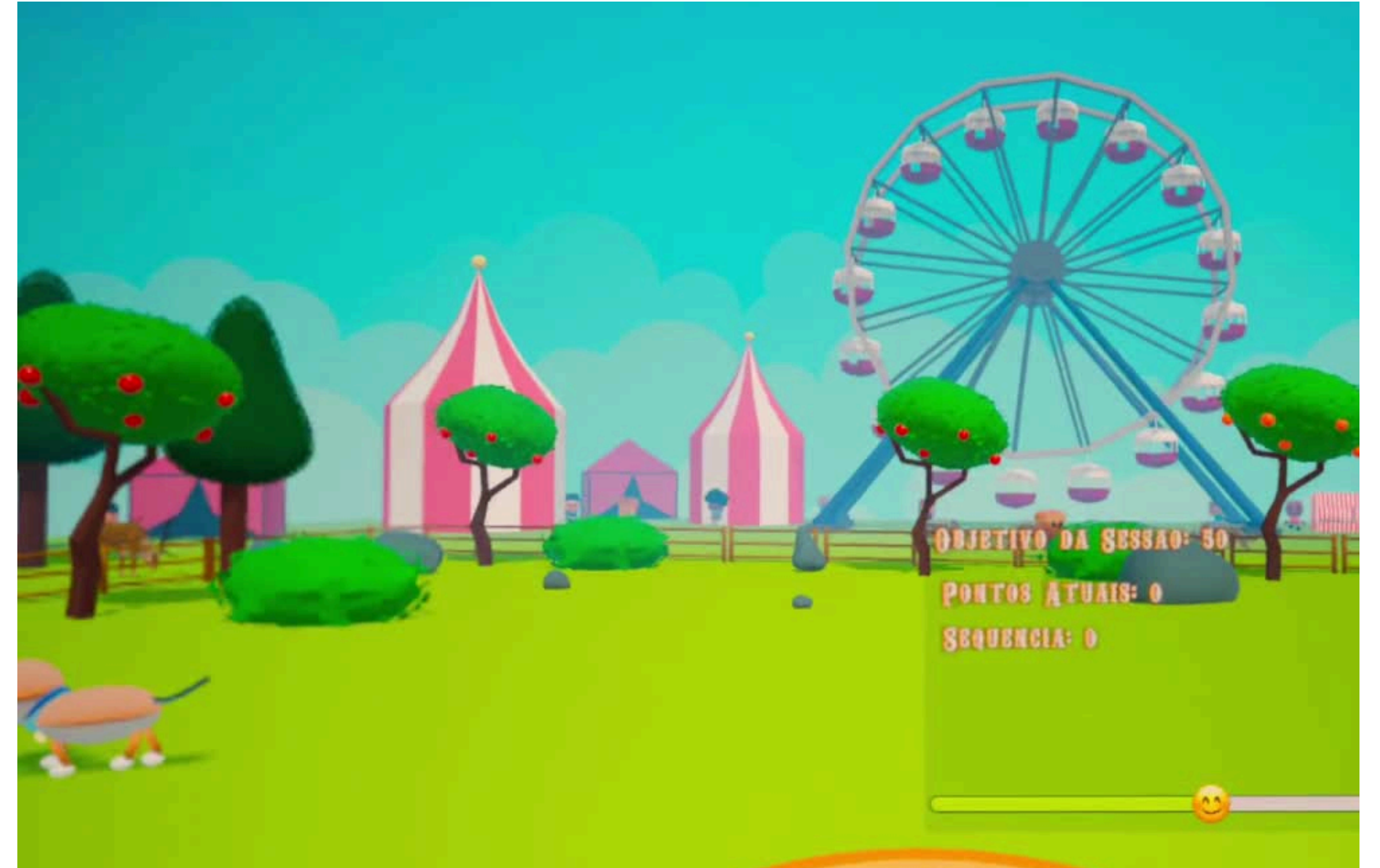
- Throw frisbee for a virtual dog
- elbow extension and wrist coordination
- Counters flexor synergy pattern

Cognitive

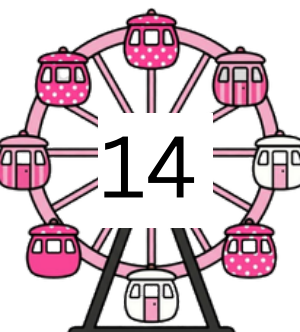
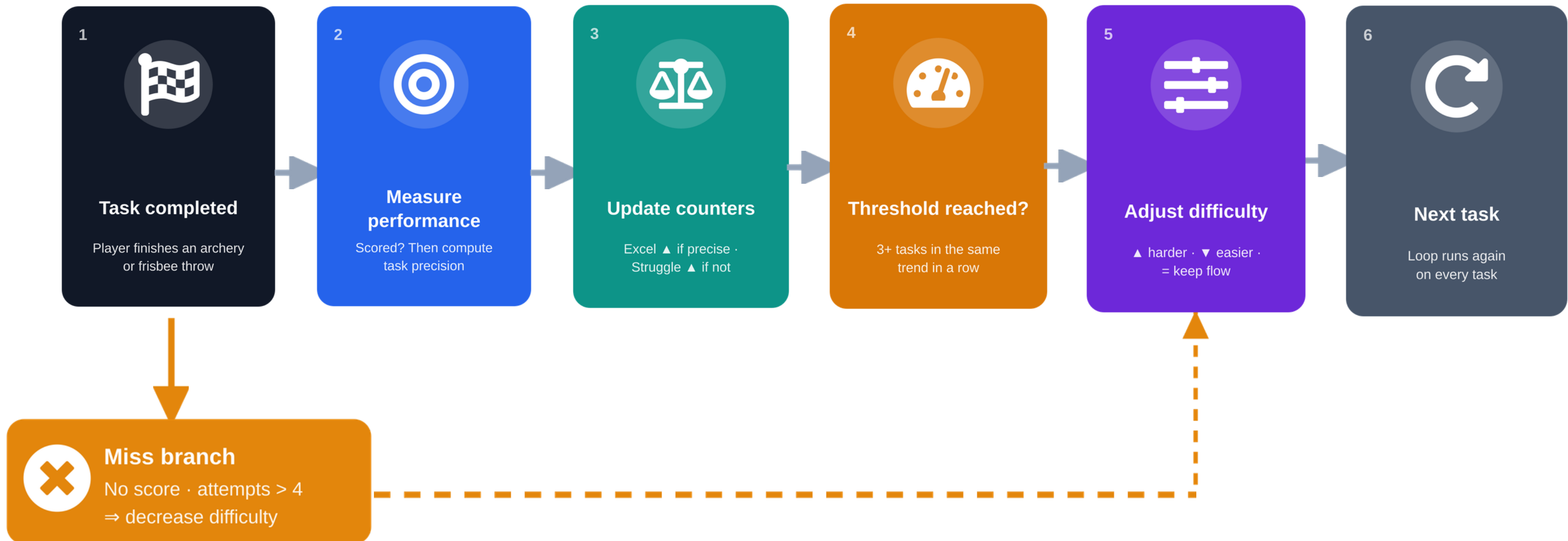
- Visuospatial integration and reaction capability

Personalization Parameters

- Target quantity, distance, movement and visibility



Personalization Algorithm



User Studies

- Wave 1: informal testing, 6–8 participants issue identification
- Wave 2: formal study, 38 participants (ages 21–57)
- Parallel: emotion module study (17 participants)



User Study Wave 2

Objectives

- Formally evaluate prototype usability with 38 participants
- Assess automatic personalization perceptibility by participants
- Validate interaction clarity, comfort and immersion

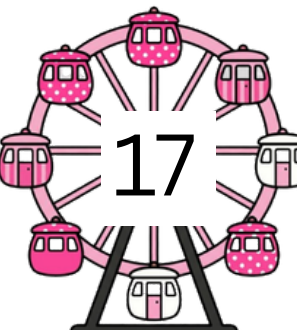
Evaluation Instruments

- Post-session Likert questionnaire with System Usability Scale (SUS)



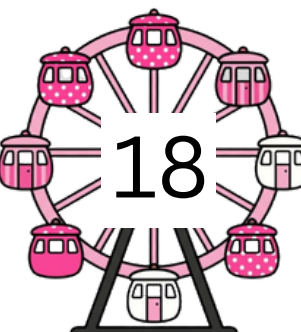
Results

- SUS: 82/100 — Excellent (Bangor et al., 2009)
- 31/38 (82%) perceived automatic personalization, 33/38 (87%) felt the system adapted
- 37/38 (98%) found the experience motivating
- 26/38 (68%) asked to keep playing



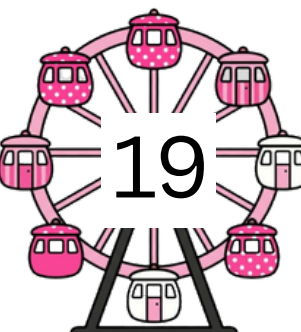
Limitations

- Only one personalization method was used
- Requires stroke survivors assessment
- Needs validation with longitudinal study



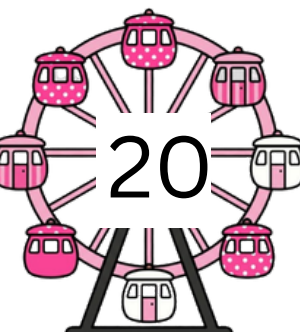
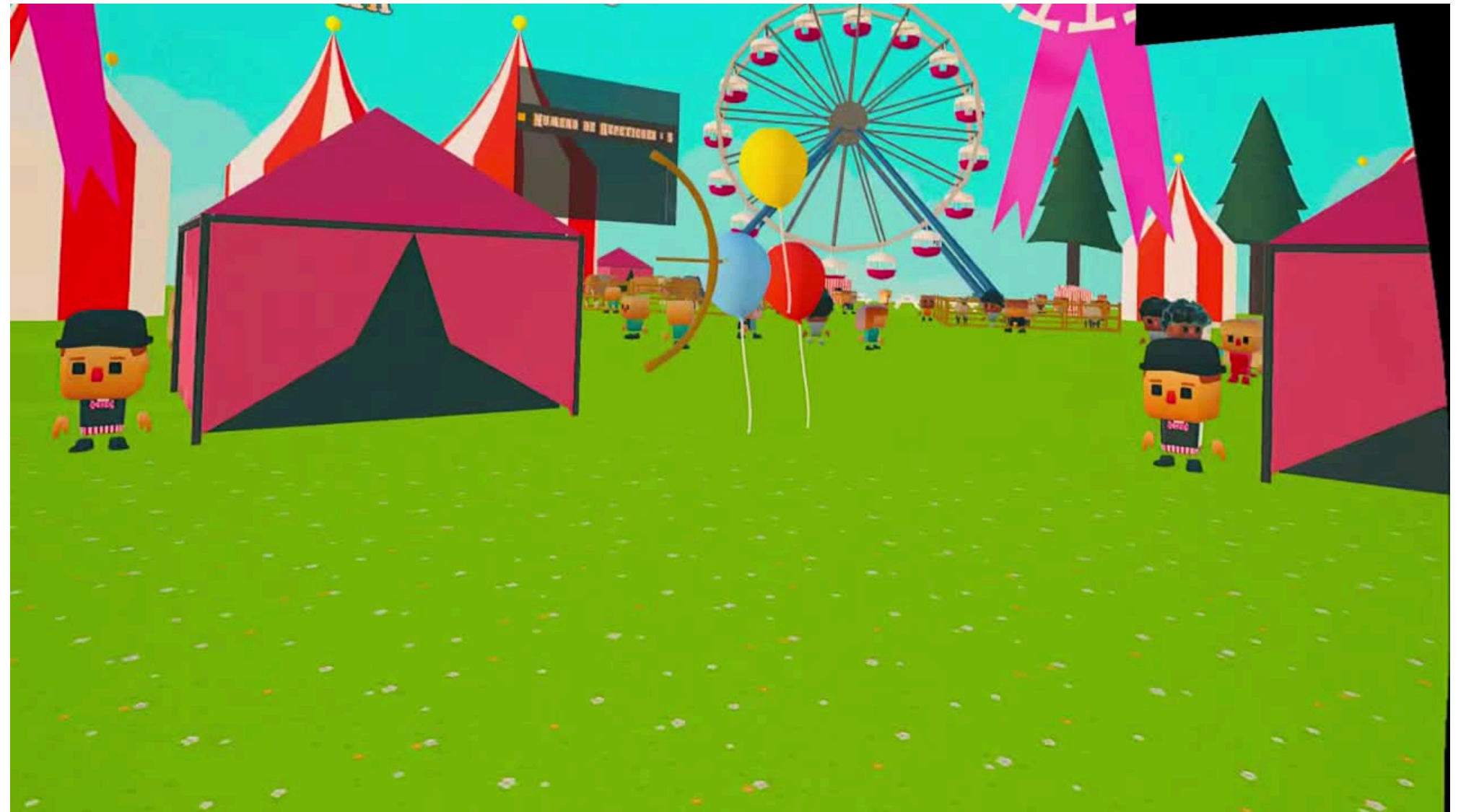
Conclusion

- Strong usability, motivation & perceptible personalization validated
- Automatic personalization confirmed functional by user
- Importance with HCD methodology



Future Work

- Expand mini-game library
- Configurable session protocol



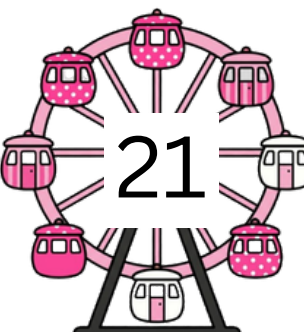
Contributions

Techinal

- Emotion Recognition Module integrated into Country Fair VR

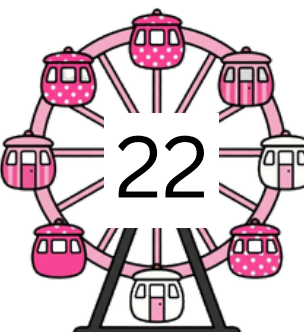
Papers

- Figueiredo et al. (2026) : Addressing Personalized Virtual Reality for Stroke Rehabilitation Through a Country Fair Serious Game Approach: A Proposal, IEEE VRW
- Figueiredo et al. (2026) : Comparing Methods for Integrating Personalized eXtended Reality in Stroke Rehabilitation, ECSCW
- Guerreiro, Palricas et al. (2026) : Visualization Framework for Shared Emotional Awareness in Personalized VR for Stroke Recovery, IV
- Palricas et al. (under review): Exploring a Virtual Pet as an adaptive agent in personalized VR rehabilitation: A Country Fair Serious Game for Stroke Survivors, ISMAR



Acknowledgements

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Any Questions ?



Thank you for your attention!

